

Power Your Career with Power Electronics

With so many developments taking place on the technology front, job seekers can expect a phenomenal increase in opportunities in power electronics. At present, there is a shortage of trained manpower, which is why experienced power electronics engineers are in high demand



■ MONIKA BHATI

India's power generation capacity is on pace to more than double in the next few years. As the country focuses on the power sector to continue its growth trajectory, there are a number of exciting opportunities waiting for you in the field of power electronics.

The Ministry of Power is actively pushing for 'Power for All by 2012'—a mission that would require our installed power generation capacity to reach at least 200,000 MW by 2012 from the present level of 114,000 MW. Power is identified as one of those sec-

tors which hold a lot of promise for growth and expansion. An important ingredient for the rapid growth of the economy, power has a set of challenges as well as opportunities for the entire energy value chain in India.

The 'Power for All by 2012' target in the current plan period requires active participation of all stakeholders. Digging down the power field, you can also become an active part of our nation-building story and a sector that will provide immense scope and opportunity.

Dalip Sharma, managing director, Delta-India and SAARC, shares, "The unprecedented growth in the Indian

electronics demand has spawned a corresponding spurt in the domestic power electronics industry. The power backup (including batteries) is a \$1.1-billion revenue industry in India. Also, there is a shift towards 'develop and make' model vis-à-vis the previous 'buy' and 'buy-and-make' policy. Owing to the presence of global electronics manufacturing services (EMS) majors in India and their plans for increased investments with more outsourcing of manufacturing by both Indian and global original equipment manufacturers, opportunities for the Indian talent pool are also increasing and the growth opportunities seem to be quite promising."

Power electronics has a very high potential as now "it has shifted from a mere IT product (UPS, inverter, etc) to powering the whole plant and also other sectors like lifts and colour photo machines," adds R.K. Bansal, founder and managing director, Uniline Energy Systems.

A brief on power electronics

Power electronics is the application of solidstate electronics for the control and conversion of electric power. The task of power electronics is to process and control the electric energy flow by supplying voltages and currents in a form optimally suited for user loads.

Power conversion systems can be categorised based on the type of the input and output power: AC-to-DC (rec-